

MINIMAL MODELS IN NUMERICAL DIMENSION ONE

OPENAI

ABSTRACT. We resolve the numerical-dimension-one case of the minimal-model conjecture for smooth connected complex projective varieties of dimension at least three. If K_X is pseudo-effective and $\kappa_\sigma(X, K_X) = 1$, with κ_σ defined by section growth with a fixed ample twist, then X admits a projective $\mathbb{Q}$ -factorial terminal minimal model.

1. INTRODUCTION

The minimal-model problem asks whether a smooth projective variety whose canonical divisor is pseudo-effective admits a birational model with nef canonical divisor. Here pseudo-effective means that the numerical class lies in the closure of the cone generated by effective divisors, and nef means nonnegative degree on every integral curve. The models allowed by the minimal model program are normal and may have terminal singularities. A minimal model is required to improve canonical discrepancies and to extract no divisors, as well as to have nef canonical divisor. These requirements retain the birational information of the original variety; the precise comparison used here is part of Theorem 1.1.

We prove existence in the case where the canonical divisor has numerical dimension one, using its growth of sections with a fixed ample twist. The argument begins with minimal model programs for positive boundary perturbations, but its final step is a criterion on surfaces. In particular, the proof does not require a general minimal-model existence theorem.

1.1. The invariant and the main theorem. For a Cartier divisor D on a smooth projective n -fold X , we use Nakayama's section-growth invariant [Nak04] in the following form:

$$(1.1) \quad \kappa_\sigma(X, D) = \max \left\{ k \in \{0, \dots, n\} : \limsup_{m \rightarrow \infty} \frac{h^0(X, \mathcal{O}_X(mD + A))}{m^k} > 0 \right\}.$$

The maximum is $-\infty$ if the set is empty, and m ranges over positive integers. The twist A is fixed before taking the limsup. Thus $\kappa_\sigma(X, D) = 1$ rules out positive quadratic limsup for *every* fixed ample Cartier twist. We will use exactly this consequence, including when quadratic growth is obtained only along the multiples of one fixed positive integer. It does not assert an upper bound of the form $h^0(X, mD + A) = O(m)$.

This growth convention should be distinguished from the intersection formula for a nef divisor,

$$\nu(D) = \max\{k \in \{0, \dots, n\} : D^k \cdot H^{n-k} > 0\},$$

where H is ample. We do not use that formula for K_X before constructing a nef model, nor do we need a comparison theorem between numerical-dimension conventions.

Theorem 1.1. *Let X be a smooth connected complex projective variety of dimension $n \geq 3$. Suppose that K_X is pseudo-effective and $\kappa_\sigma(X, K_X) = 1$, with the convention (1.1). Then there are a normal projective $\mathbb{Q}$ -factorial terminal variety Y with K_Y nef and a finite sequence $\phi : X \dashrightarrow Y$ of K -negative divisorial contractions and flips. The inverse of ϕ contracts no prime divisor. On a common smooth projective resolution $p : W \rightarrow X$, $q : W \rightarrow Y$, compatible canonical divisors satisfy*

$$(1.2) \quad p^*K_X = q^*K_Y + E,$$

where E is an effective q -exceptional $\mathbb{Q}$ -divisor whose support contains the strict transform of every prime divisor of X contracted by ϕ .

The theorem gives a positive resolution of the numerical-dimension-one case of the minimal-model conjecture for smooth complex projective varieties of dimension at least three. In particular it applies to smooth fivefolds, with no hypothesis on $\chi(X, \mathcal{O}_X)$.

A good minimal model additionally requires a positive multiple of K_Y to be generated by global sections. This is the semiample conclusion of abundance. Nonvanishing asks for a nonzero section of a positive multiple of the canonical divisor. Neither conclusion is asserted by Theorem 1.1 alone; the fixed ample twist in (1.1) is retained throughout the section-growth argument. The following separate consequence uses log abundance.

Corollary 1.2 (Good minimal model). *Under the hypotheses of Theorem 1.1, the same $\mathbb{Q}$ -factorial terminal endpoint Y has semiample K_Y ; hence Y is a good minimal model of X .*

Proof. Apply the log abundance theorem [Ope26, Theorem 1.1] to the projective lc pair $(Y, 0)$ over $\mathbb{C}$, whose $\mathbb{Q}$ -Cartier canonical divisor is nef. $\square$

1.2. The problem and the methods in the literature. The three-dimensional program developed through work of Reid, Benveniste, Kawamata, Shokurov, Kollár, Miyaoka and others, with Mori’s flip theorem providing a central step in minimal-model existence [Mor88]; see [KM98] for the broader development. The minimal model program replaces a variety by successive contractions and flips along canonical-negative extremal rays; the standard foundations, including the singularities and discrepancy comparisons used here, are presented in Kollár–Mori [KM98]. Birkar’s formulation for log pairs [Bir10, Conjecture 1.1] places existence of log minimal models alongside the Mori fiber space alternative. A decisive advance was the work of Birkar, Cascini, Hacon and McKernan [BCHM10], which established minimal-model existence for varieties of general type and finite minimal model programs with scaling for klt pairs with big boundary. The positive perturbations in our proof lie within precisely this established range. Their boundary transforms remain big; they need not remain ample.

At the other end of the numerical spectrum, Nakayama developed the section-growth invariants and divisorial Zariski decompositions that underlie numerical approaches to the canonical divisor [Nak04]. Druel proved termination results for directed programs in numerical dimension zero [Dru11, Corollary 3.4], and Gongyo extended this method to prove minimal-model existence for $\mathbb{Q}$ -factorial dlt pairs of numerical log Kodaira dimension zero [Gon11, Theorem 1.1]. The problem here is to pass from arbitrarily small positive perturbations to the canonical divisor in the next numerical dimension. A finite program for each positive parameter does not by itself terminate their concatenation at parameter zero.

Our first reduction also has a specific methodological predecessor. Alexeev, Hacon and Kawamata use fourth homology to control certain four-dimensional flips [AHK07, Lemma 3.1 and its proof]. We use the same kind of decrease after a discrepancy argument has removed codimension-two components on the flipped side. In arbitrary dimension n , the relevant finite-dimensional space is the span of algebraic $(n - 2)$ -cycle classes in degree $2n - 4$. The proof of

Lemma 2.3 gives the needed statement directly; the four-dimensional result is a precedent for the method, not an imported termination theorem in all dimensions.

The second reduction combines two established strands of positivity. Nadel's multiplier-ideal vanishing theorem [Nad90] supplies the vanishing used to extend sections from a very general surface while allowing the perturbation to shrink with the section degree; we use the formulation in Lazarsfeld [Laz04b, Theorem 9.4.8], [Laz10, Theorem 2.4]. Surface Riemann–Roch then converts positive square into quadratic section growth. A different surface, through a point of a hypothetical negative curve, turns a vanishing-order estimate into a fixed exceptional divisor. The Hodge index theorem makes its square uniformly negative. The surface intersection calculations are classical [Har77, Chapter V, Section 1]; our use of them keeps one surface resolution and one exceptional curve fixed while the rational perturbations vary. This uniformity is what produces a contradiction.

For comparison with results concerning sections of the unperturbed canonical divisor, Lazić and Peternell prove nonvanishing for an *already minimal* projective terminal variety with numerical dimension one and nonzero Euler characteristic [LP18, Theorem 6.7]. Liu and Xu prove existence of a good minimal model for smooth projective varieties of dimension at most five and numerical dimension at most one, assuming nonnegative Kodaira dimension [LX25, Theorem 1.2]. They also obtain a good minimal model for a smooth projective variety of dimension at most four with nonzero Euler characteristic and $0 \leq \kappa_\sigma(X, K_X) \leq 1$ [LX25, Corollary 5.2]. These hypotheses and conclusions differ from the existence statement above. We do not use their nonvanishing or abundance results as proof inputs.

1.3. The route through the proof. The first step is to study a sequence of shrinking perturbations on one fixed model. Starting with an effective ample rational boundary B , we run finite minimal model programs for decreasing positive coefficients of its transform. Every step is K -negative. Discrepancy and cycle-class arguments show that only finitely many steps change a locus of codimension at most two; they do not assert termination of the concatenated programs. Proposition 2.1 therefore supplies a finite canonical prefix $X \dashrightarrow Y$ such that multiples of

$$M_j = K_Y + t_j B_Y, \quad t_j > 0, \quad t_j \rightarrow 0,$$

are generated outside closed subsets $Z_j \subset Y$ of codimension at least three. Here B_Y is the transform of B ; the generated multiple and the bad set may depend on j .

Two surfaces then show that K_Y is nef. Fix a very ample divisor H on Y . A very general complete intersection surface S avoids every Z_j and the singular locus. The restrictions $M_j|_S$ are semiample, so their limit $K_Y|_S$ is nef. Lemma 3.1 uses multiplier-ideal vanishing to extend sections from this fixed surface with one fixed twist. Positive square $K_Y^2 \cdot H^{n-2}$ would consequently give quadratic section growth. Section 4 transfers that growth to the original X , where $\kappa_\sigma(X, K_X) = 1$ rules it out. Thus the square is zero.

This first surface need not meet a curve of negative K_Y -degree. To exclude such a curve, including one in the singular locus, take another complete intersection from $|H|$ through one of its points, and choose a surface component through that point. On every component the restriction of M_j has nonnegative square; these squares, counted with multiplicity, sum to $M_j^2 \cdot H^{n-2} \rightarrow 0$. Thus the square on the chosen surface also tends to zero. For all sufficiently small perturbations, an ambient jet estimate on a resolution of Y forces sections of Cartier multiples kM_j to vanish to order at least ck at one fixed point above the curve, with $c > 0$ independent of j and k . On a fixed resolution of the second surface, this produces an exceptional fixed part. The Hodge index theorem bounds its square above by $-\epsilon k^2$, with $\epsilon > 0$ fixed, while the residual moving system has nonnegative square. Thus the perturbations have uniformly positive square on that surface, contradicting its zero-square limit. This is the nefness criterion of Proposition 3.4. The

finite prefix already supplies the remaining birational and canonical-comparison assertions of Theorem 1.1.

1.4. Conventions. All varieties and morphisms are over $\mathbb{C}$, and all birational models in the proof are projective. We use compatible canonical divisors. Log discrepancies are normalized by

$$a(F; U, \Delta) = 1 + \text{coeff}_F(K_{\tilde{U}} - r^*(K_U + \Delta)),$$

where $r : \tilde{U} \rightarrow U$ is a resolution carrying the prime divisor F . Terminality at zero boundary means $a(F; U, 0) > 1$ for every exceptional prime F over U ; an original prime divisor has log discrepancy 1. For an effective rational boundary, klt means that all log discrepancies are positive [KM98, Chapter 2].

A birational map *extracts no divisors* if its inverse contracts no prime divisor. Intersection numbers of rational Cartier divisors are defined after clearing denominators. On an integral surface that need not be normal, we use Cartier intersections with its fundamental cycle [Ful98, Chapter 2]. Characteristic-zero projective resolutions and common resolutions are used throughout [Hir64].

2. POSITIVE PERTURBATIONS ON A FIXED MODEL

The construction in this section does not use the numerical-dimension hypothesis. It provides one model on which small positive perturbations of the canonical divisor have base locus of codimension at least three.

Proposition 2.1. *Let X be a smooth projective variety of dimension $n \geq 3$ with pseudo-effective K_X . There are an effective ample $\mathbb{Q}$ -divisor B on X , an integer $s \geq 0$, and a finite sequence of K -negative divisorial contractions and K -flips*

$$X \dashrightarrow Y$$

with the following properties.

- (i) Y is projective, $\mathbb{Q}$ -factorial, and terminal. The map $X \dashrightarrow Y$ extracts no divisors.
- (ii) Put $L = K_Y$, let B_Y be the transform of B , and set

$$t_j = 2^{-j}, \quad M_j = L + t_j B_Y \quad (j > s).$$

For each $j > s$ there is a closed subset $Z_j \subset Y$ of codimension at least three such that $|kM_j|$ is defined by a Cartier divisor and is base point free on $Y \setminus Z_j$ for every sufficiently divisible positive integer k .

- (iii) *On a common smooth projective resolution $p : W \rightarrow X$, $q : W \rightarrow Y$, compatible canonical divisors satisfy*

$$(2.1) \quad p^*K_X = q^*K_Y + E,$$

where $E \geq 0$ is q -exceptional and its support contains the strict transform of every prime divisor of X contracted by $X \dashrightarrow Y$.

Here and below, “sufficiently divisible” allows the required divisor index to depend on j . No uniform Cartier index for an infinite sequence of models will be needed.

2.1. Discrepancies and codimension two. Write $a(F, U) = a(F; U, 0)$ with the normalization fixed in the introduction. We record the strictness statement needed for the finiteness argument.

Lemma 2.2. *Suppose that $U \dashrightarrow U^+$ is a K -negative divisorial contraction or a K -flip between normal projective $\mathbb{Q}$ -Gorenstein varieties. Write $h : U \rightarrow R$ for the contraction and $h^+ : U^+ \rightarrow R$*

for the other morphism, with h^+ the identity in the divisorial case. For every prime divisor F over these varieties,

$$a(F, U) \leq a(F, U^+).$$

The inequality is strict if the center of F on R is contained in the locus where either h or h^+ has a positive-dimensional fiber.

Proof. Take a common smooth projective resolution carrying F :

$$\begin{array}{ccccc} & & W & & \\ & p \swarrow & & \searrow q & \\ U & \xrightarrow{\quad \text{---} \quad} & & & U^+ \\ & h \searrow & & \swarrow h^+ & \\ & & R & & \end{array}$$

Write $g = h \circ p = h^+ \circ q : W \rightarrow R$ and set $D = p^*K_U - q^*K_{U^+}$. This divisor is q -exceptional, since the step extracts no divisors. On every curve contracted by g , the divisors $-p^*K_U$ and $q^*K_{U^+}$ have nonnegative degree; in the divisorial case the second has degree zero. Thus $-D$ is g -nef, and hence q -nef. The negativity lemma gives $D \geq 0$; see [KM98, Lemmas 3.38–3.39].

To verify the support assertion, let $r \in R$ be a closed point with a positive-dimensional fiber on either side. There is a curve in $g^{-1}(r)$ mapping onto a curve in that fiber: take a component dominating that curve and cut it by sufficiently general ample divisors. The relative ampleness signs give $D \cdot C < 0$ for this lifted curve. Consequently $g^{-1}(r)$ meets $\text{Supp } D$.

We use the fiber-support form of the negativity lemma [KM98, Lemma 3.39]: the support of an effective divisor antinef over a proper birational morphism to a normal variety either contains or is disjoint from each fiber. In the present setting it can also be seen directly on the reduced irreducible components of the fiber. If such a component T is not contained in $\text{Supp } D$ but meets it, a Cartier multiple of D restricts to a nonzero effective Cartier divisor on the integral projective variety T . Its degree against sufficiently many ample hyperplanes is positive, producing a curve contracted by g with positive D -degree. This contradicts antinefness. This argument does not require T to be normal or the fiber to be reduced: only its support is at issue. Components of dimension zero already lie in the support if they meet it. The fibers of g are connected because R is normal and g is proper and birational [Har77, Corollary III.11.4]. Containment therefore propagates through their reduced irreducible components.

It follows that every fiber in question is contained in $\text{Supp } D$. If the center of F on R is contained in this locus, then $F \subset \text{Supp } D$ on our chosen resolution. Finally,

$$a(F, U^+) - a(F, U) = \text{coeff}_F D,$$

which proves both assertions. $\square$

Lemma 2.3 (Stabilization in codimension two). *Let*

$$X = U_0 \dashrightarrow U_1 \dashrightarrow U_2 \dashrightarrow \cdots$$

be a finite or infinite sequence of K -negative divisorial contractions and K -flips of projective $\mathbb{Q}$ -factorial varieties, starting from a smooth n -fold, $n \geq 3$. After finitely many steps, each remaining map is an isomorphism outside closed subsets of codimension at least three on both sides.

Proof. The usual MMP properties give preservation of $\mathbb{Q}$ -factoriality, invariance of the Picard number under a flip, and a drop of one under an elementary divisorial contraction [KM98, Propositions 3.36–3.37]. There are therefore only finitely many divisorial contractions. Since the steps extract no divisors, only finitely many prime divisors of X can disappear; denote this finite set by $\mathcal{D}$.

All models are terminal. This follows from [KM98, Corollaries 3.42–3.43], or directly from Lemma 2.2: previously exceptional valuations retain log discrepancy greater than one, and a newly contracted divisor increases strictly from log discrepancy one. Moreover, every valuation exceptional over the smooth X has integral log discrepancy at least two. Discrepancies never decrease along the sequence.

The flipped loci. Consider a flip whose target has a codimension-two component T of its flipped locus. A terminal variety is smooth in codimension two [KM98, Corollary 5.18]. Blowing up T at its generic smooth point therefore defines a prime valuation F with

$$a(F, U_{i+1}) = 2.$$

Its center on the contraction base is contained in the locus of positive-dimensional fibers of the flipped contraction. Lemma 2.2 implies

$$(2.2) \quad a(F, X) \leq a(F, U_i) < 2 = a(F, U_{i+1}).$$

Thus F is not exceptional over X : it is an original prime divisor. It is exceptional over U_{i+1} , so it belongs to $\mathcal{D}$. After this occurrence its discrepancy is at least two forever, and it cannot satisfy (2.2) at a later step. Each flip with such a target component therefore uses a distinct member of a finite set. After a finite prefix, all steps are flips and every flipped locus has codimension at least three.

The flipping loci. We have removed codimension-two components on the target side. It remains to remove them on the source side; only then can a general surface be transported unchanged through every later finite stage. The argument adapts the homological decrease used in the proof of [AHK07, Lemma 3.1], keeping only classes of algebraic cycles. Put $d = n - 2$ and define the finite-dimensional real vector space

$$\mathcal{A}_d(V) = \text{span}_{\mathbb{R}}\{[T] : T \subset V \text{ is a closed integral } d\text{-fold}\} \subset H_{2d}(V, \mathbb{R}).$$

Cycle classes, Borel–Moore localization, and their compatibility with restriction to an open subset are as in [Ful98, Section 19.1]. The homology of the compact complex algebraic varieties here is finite-dimensional and vanishes above their real dimension.

For a remaining flip $U \dashrightarrow U^+$ over R , remove the images in R of both exceptional loci and take inverse images. This gives a common open subset $\mathcal{U}$. Indeed, a proper birational morphism to a normal variety is an isomorphism over its quasi-finite locus. Each exceptional locus consists of positive-dimensional fibers; its image has dimension at most one less than its own. Since the source contraction is small and the target exceptional locus has dimension at most $n - 3$, the closed complements $Z \subset U$ and $Z^+ \subset U^+$ satisfy

$$\dim Z \leq d, \quad \dim Z^+ \leq d - 1.$$

The additional points removed outside the exceptional loci lie in an isomorphism locus and have dimension at most $n - 3$.

Restriction gives maps

$$\mathcal{A}_d(U) \longrightarrow H_{2d}^{\text{BM}}(\mathcal{U}, \mathbb{R}) \longleftarrow \mathcal{A}_d(U^+)$$

with the same image: intersect an integral d -fold with $\mathcal{U}$ and take its closure on the other model, while a d -fold contained in the complement restricts to zero. The right-hand map is injective, by the localization sequence and $H_{2d}(Z^+, \mathbb{R}) = 0$. Hence

$$(2.3) \quad \dim \mathcal{A}_d(U^+) \leq \dim \mathcal{A}_d(U).$$

If the flipping locus has a d -dimensional component T , its class belongs to the kernel of the left-hand restriction and is nonzero: for an ample Cartier divisor A ,

$$\langle c_1(A)^d, [T] \rangle = A^d \cdot T > 0.$$

Then (2.3) is strict. A nonnegative integer can drop only finitely many times, so eventually the flipping loci also have codimension at least three. The same construction of the common open then gives this codimension bound on both complements. $\square$

2.2. Construction of the perturbations.

Proof of Proposition 2.1. Choose an effective ample rational divisor B for which (X, B) is klt and $K_X + B$ is ample. For example, take a sufficiently positive general smooth very ample divisor and multiply it by a rational number strictly between zero and one. Since K_X is pseudo-effective, $K_X + tB$ is big for every rational $t > 0$.

We shall repeatedly use a simple consequence of normality. If a birational map $X \dashrightarrow U$ extracts no divisors, every prime divisor of U corresponds to a prime divisor of X . For compatible transformed divisors, the divisorial criterion for regularity of a rational section consequently gives

$$(2.4) \quad H^0(X, kD) \hookrightarrow H^0(U, kD_U)$$

whenever the multiples are Cartier. Thus effectiveness and bigness of B_U , and bigness of $K_U + tB_U$, persist on every model constructed below. In particular, we do not need B_U to be ample.

Finite stages. Set $t_j = 2^{-j}$ for every integer $j \geq 0$. Starting with $V_0 = X$, construct V_{j+1} from V_j by running the MMP for

$$K_{V_j} + \Delta_j, \quad \Delta_j = t_{j+1}B_{V_j},$$

with scaling of $C_j = (t_j - t_{j+1})B_{V_j}$. The precise input is [BCHM10, Corollary 1.4.2]: for a projective $\mathbb{Q}$ -factorial klt pair (V, Δ) with big boundary Δ , and $C \geq 0$ such that $(V, \Delta + C)$ is klt and $K_V + \Delta + C$ is nef, the MMP with scaling of C terminates. Here the inductive hypothesis is that $(V_j, t_j B_{V_j})$ is klt and its adjoint is nef. The smaller pair is klt because B_{V_j} is effective; its boundary is big by (2.4); and the larger pair is exactly the one in the inductive hypothesis.

The adjoint remains big by (2.4), so a Mori fiber space cannot occur: an effective representative of a positive multiple cannot have negative degree on curves covering general fibers. Each stage is consequently a finite, possibly empty, sequence ending at a nef adjoint. Standard MMP preservation gives the next klt pair and a projective $\mathbb{Q}$ -factorial model. The rational divisor $K_{V_j} + t_j B_{V_j}$ is nef and big, hence semiample by the klt base point free theorem [KM98, Theorem 3.3].

Canonical steps. To apply Lemma 2.3, we must identify every adjoint step as a canonical MMP step. This requires canonical negativity on the contracted ray and, for a flip, canonical ampleness on the flipped side. On its source U , let $D = K_U + t_{j+1}B_U$. The contracted ray R_0 satisfies

$$D \cdot R_0 < 0, \quad (D + \lambda C) \cdot R_0 = 0, \quad \lambda > 0,$$

where $C = (t_j - t_{j+1})B_U$ is the current scaling divisor. Therefore $B_U \cdot R_0 > 0$ and $K_U \cdot R_0 < 0$. This is the usual ray calculation for a directed program [BCHM10, proof of Corollary 1.3.3], and it already identifies each divisorial step as a K -negative divisorial contraction.

For a small contraction $h : U \rightarrow R$, the adjoint flip makes D^+ relatively ample. We show that K_{U^+} is relatively ample as well. Choose the positive rational number

$$b = \frac{K_U \cdot R_0}{D \cdot R_0}.$$

The divisor $K_U - bD$ is numerically trivial over R and descends up to rational linear equivalence. To see this, a Cartier multiple is relatively semiample by the relative klt base point free theorem for the pair $(U, t_{j+1}B_U)$ [KM98, Theorem 3.24]: subtracting its adjoint D gives relative ampleness, because $-D$ is relatively ample. Its semiample morphism is constant on each connected projective fiber, as its degree on every fiber curve is zero. The resulting image is proper and quasi-finite

over R , because each connected fiber maps to one point. It is therefore finite and birational over the normal R , hence equals R . Thus $K_U - bD \sim_{\mathbb{Q}} h^*A$ for a rational Cartier divisor A on R .

Both morphisms in the adjoint flip are small, so this relation transforms to

$$K_{U^+} - bD^+ \sim_{\mathbb{Q}} (h^+)^*A.$$

Since D^+ is relatively ample and $b > 0$, so is K_{U^+} . The adjoint flip is therefore a K -flip. Thus all steps of the finite stages satisfy the hypotheses of Lemma 2.3.

The fixed model. Concatenate the finite stages and apply Lemma 2.3. Choose an endpoint $Y = V_s$ after the resulting finite prefix. If only finitely many nonempty stages occur, choose s after the last one. For every $j > s$, the finite composition $Y \dashrightarrow V_j$ is an isomorphism outside codimension-at-least-three closed subsets on both models. This property is preserved under finite composition: on a common open, remove the next bad set and take its closure on the preceding model; its dimension does not increase. Let Z_j be the resulting closed subset of Y .

The divisor $M_j = K_Y + t_j B_Y$ agrees on this common open with $K_{V_j} + t_j B_{V_j}$. Choose a multiple Cartier on both varieties and base point free on V_j . A line bundle on a normal variety has the same global sections after deleting a subset of codimension at least two, since it is reflexive. The two spaces of sections are therefore identified, and the complete system on Y is generated outside Z_j . This proves (ii) for each j .

Finally, Y is projective, $\mathbb{Q}$ -factorial and terminal by the preceding construction, and a finite composition of these steps extracts no divisors. On a common smooth projective resolution of the finite chain, the effective differences in Lemma 2.2 telescope to (2.1). A divisor mapping onto a prime divisor of Y also corresponds to a prime divisor of X , so its coefficient in E is zero. Thus E is q -exceptional. For an original prime divisor contracted on Y , terminality gives log discrepancy greater than one on Y , compared with one on X . Its strict transform has positive coefficient in E , proving (iii). $\square$

3. TWO SURFACE TESTS FOR NEFNESS

Our goal is a numerical criterion for nefness. The first surface will convert positive intersection square into section growth with one fixed twist. The second surface will show that a negative curve forces that square to be positive. Together these statements let the numerical-dimension hypothesis, imposed later on the original variety, rule out every negative curve. The arguments apply to rational divisors on an n -dimensional variety, for every $n \geq 3$. We use the following precise approximation hypothesis:

$$(3.1) \quad \begin{aligned} &Y \text{ is a normal projective } n\text{-fold over } \mathbb{C}, \quad n \geq 3, \quad \text{codim}_Y \text{Sing } Y \geq 3, \\ &L, B_Y \text{ are } \mathbb{Q}\text{-Cartier}, \quad t_j \in \mathbb{Q}_{>0}, \quad t_j \longrightarrow 0, \quad M_j = L + t_j B_Y, \\ &\text{for each } j \text{ there are a closed } Z_j \subset Y, \quad \text{codim}_Y Z_j \geq 3, \quad \text{and } d_j \in \mathbb{Z}_{>0} \text{ such that} \\ &d_j M_j \text{ is Cartier and } |d_j M_j| \text{ is generated by global sections on } Y \setminus Z_j. \end{aligned}$$

Every positive multiple of d_j has the same generation property. The integers d_j may depend on j . Restrictions of rational Cartier divisors below mean restrictions as rational line bundles. Neither effectiveness nor positivity of B_Y is part of (3.1).

3.1. A fixed twist detects positive surface square.

Lemma 3.1 (Surface growth with a fixed twist). *Assume (3.1), and let H be a very ample Cartier divisor on Y . Fix a projective resolution $\pi : \hat{Y} \rightarrow Y$ which is an isomorphism over the smooth locus. There are a smooth complete intersection surface $S \subset Y$, cut out by $n - 2$ members*

of $|H|$, and a Cartier divisor P on $\widehat{Y}$ with the following properties. Identify S with its inverse image in $\widehat{Y}$. Then $L|_S$ is nef, so

$$(3.2) \quad L^2 \cdot H^{n-2} \geq 0.$$

If dL is Cartier, then for every positive integer m divisible by d the restriction map

$$(3.3) \quad H^0(\widehat{Y}, m\pi^*L + P) \longrightarrow H^0(S, mL|_S + P|_S)$$

is surjective. In particular, as $m \rightarrow \infty$ through multiples of this one fixed integer d ,

$$(3.4) \quad h^0(\widehat{Y}, m\pi^*L + P) \geq \frac{m^2}{2} L^2 \cdot H^{n-2} + O(m),$$

where the error term depends only on the fixed surface and its divisors.

Proof. Put $r = n - 2$. Very general members of $|H|$ have a smooth complete intersection S avoiding $\text{Sing } Y$ and every Z_j . Indeed, each of these countably many closed sets has dimension at most $n - 3$, so the condition that r hyperplanes meet any one of them is a proper closed incidence condition. Over $\mathbb{C}$ these can be avoided simultaneously, together with the closed conditions excluded by Bertini's theorem. The inverse image of S is therefore exactly the smooth complete intersection cut out by the r pulled-back equations; there are no additional components over the singular locus. Each $M_j|_S$ is semiample, hence nef. Passing to the limit on every curve on S shows that $L|_S$ is nef and proves (3.2).

Write $A = \pi^*H$, choose an ample Cartier divisor A_0 on $\widehat{Y}$, and make the fixed choice

$$(3.5) \quad P = K_{\widehat{Y}} + rA + A_0.$$

Since A is nef and ampleness is open [Laz04a, Corollary 1.4.10 and Theorem 1.4.23], there exists $\eta > 0$ such that

$$(3.6) \quad A_0 + (r - i)A - \epsilon\pi^*B_Y \quad \text{is ample whenever } 0 \leq i \leq r \text{ and } 0 \leq \epsilon < \eta.$$

For a given positive m with $d \mid m$, first choose $j = j(m)$ so that $mt_j < \eta$, and then choose a multiple k of d_j with $k > m$. The pullback of $|kM_j|$ is generated near S . A general member D_m is smooth there, by Bertini, and

$$R_m = \frac{m}{k}D_m \geq 0, \quad R_m \sim_{\mathbb{Q}} m\pi^*M_j, \quad \mathcal{J}(R_m) = \mathcal{O}_{\widehat{Y}} \quad \text{near } S.$$

Here $\mathcal{J}(R_m)$ is the multiplier ideal; its asserted local triviality follows from $m/k < 1$ and the smoothness of D_m there.

The divisors $m\pi^*L + P - iA$ are integral Cartier divisors, and

$$m\pi^*L + P - iA - K_{\widehat{Y}} - R_m \sim_{\mathbb{Q}} A_0 + (r - i)A - mt_j\pi^*B_Y$$

is ample for $0 \leq i \leq r$. Nadel vanishing [Laz04b, Theorem 9.4.8], equivalently [Laz10, Theorem 2.4], gives

$$(3.7) \quad H^q(\widehat{Y}, \mathcal{O}_{\widehat{Y}}(m\pi^*L + P - iA) \otimes \mathcal{J}(R_m)) = 0 \quad (q > 0, 0 \leq i \leq r).$$

Set $\mathcal{F} = \mathcal{O}_{\widehat{Y}}(m\pi^*L + P) \otimes \mathcal{J}(R_m)$. Tensoring the Koszul complex of the r equations with $\mathcal{F}$ yields the exact complex

$$0 \longrightarrow \mathcal{F}(-rA) \longrightarrow \cdots \longrightarrow \mathcal{F}(-A)^{\oplus r} \longrightarrow \mathcal{F} \longrightarrow \mathcal{O}_S(mL|_S + P|_S) \longrightarrow 0.$$

At points of S , the equations form a regular sequence and $\mathcal{F}$ is locally free. Outside S , one equation is a unit, so the Koszul complex is contractible even after tensoring with an arbitrary sheaf. This proves exactness everywhere. Splitting the complex into short exact sequences and using (3.7) gives surjectivity on global sections onto its last term. The inclusion $\mathcal{F} \subset \mathcal{O}_{\widehat{Y}}(m\pi^*L + P)$ then proves (3.3).

Adjunction and (3.5) give $P|_S - K_S = A_0|_S$. Hence $mL|_S + P|_S - K_S$ is ample. Kodaira vanishing, the zero-boundary ample case of [Laz10, Theorem 2.4], and surface Riemann–Roch [Har77, Chapter V, Section 1] give

$$\begin{aligned} h^0(S, mL|_S + P|_S) &= \frac{m^2}{2}(L|_S)^2 + \frac{m}{2}L|_S \cdot (2P|_S - K_S) \\ &\quad + \frac{1}{2}P|_S \cdot (P|_S - K_S) + \chi(S, \mathcal{O}_S). \end{aligned}$$

If S is disconnected, the formula is read componentwise and summed. All its coefficients are fixed before m, j, k, D_m vary. Together with (3.3), this proves (3.4). $\square$

3.2. Negative degree forces ambient vanishing. Lemma 3.1 supplies the first surface test. To obtain nefness once its square is zero, we must rule out curves missed by that very general surface, including curves in the singular locus. The next two lemmas will measure their effect on a second surface.

The next elementary estimate retains the cotangent bundle of the ambient variety. This is what permits a curve singular at the point where vanishing will be measured.

Lemma 3.2 (Uniform ambient multiplicity). *Let V be a smooth projective complex variety, let $\Gamma \subset V$ be an integral curve, and let $\iota : N \rightarrow V$ be the map from its normalization. For a section s , write $\text{ord}_z(s)$ for its order in the regular local ring of V at z . There is a positive constant g , depending only on this map, such that for every line bundle $\mathcal{D}$ on V , every nonzero section $s \in H^0(V, \mathcal{D})$, and every closed point $z \in \Gamma$,*

$$(3.8) \quad \text{ord}_z(s) \geq -\frac{\deg_N \iota^* \mathcal{D}}{g}.$$

Proof. Choose a line bundle G of positive degree on the fixed smooth curve N such that $\iota^* \Omega_V^1$ embeds as a subbundle of $G^{\oplus a}$ for some a . Such a choice follows by global generation of $(\iota^* \Omega_V^1)^* \otimes G$ for a sufficiently ample G , followed by dualization. Set $g = \deg G > 0$.

For a nonzero s , the integer

$$b = \min_{p \in \Gamma \text{ closed}} \text{ord}_p(s)$$

is finite and attained. The leading ambient jet gives a nonzero section of

$$(3.9) \quad \text{Sym}^b(\iota^* \Omega_V^1) \otimes \iota^* \mathcal{D}.$$

Here is a justification valid also at singular points of Γ . For $b \geq 1$, the bundles of principal parts on the smooth variety V have the locally split exact sequence

$$0 \longrightarrow \text{Sym}^b \Omega_V^1 \otimes \mathcal{D} \longrightarrow \mathcal{P}^b(\mathcal{D}) \longrightarrow \mathcal{P}^{b-1}(\mathcal{D}) \longrightarrow 0.$$

The lower jet of s has zero value at every closed point of the reduced curve Γ . A section of a vector bundle on a reduced variety with this property is zero, so that lower jet restricts to the zero section. The principal-parts sequence is locally split as a sequence of $\mathcal{O}_V$ -modules, and therefore remains exact after restriction to Γ and pullback to N . The restricted order- b jet consequently belongs to the left-hand bundle. Its value is nonzero at a point where the minimum b is attained; the induced map on its fiber at any point of N above that point is an isomorphism. Its pullback is thus regular and nonzero. For $b = 0$, use $s|_\Gamma$ directly. This proves (3.9) without assuming smoothness of Γ .

Taking symmetric powers of the subbundle inclusion above embeds (3.9) into a direct sum of copies of $G^{\otimes b} \otimes \iota^* \mathcal{D}$. A nonzero component is a nonzero section of this line bundle on N , so

$$0 \leq bg + \deg_N \iota^* \mathcal{D}.$$

Since $\text{ord}_z(s) \geq b$ for every $z \in \Gamma$, (3.8) follows. $\square$

3.3. A fixed exceptional curve forces a positive square.

Lemma 3.3 (Surface fixed-part estimate). *Let T be an integral projective surface, let $f : T_1 \rightarrow T$ be a projective birational morphism from a smooth surface, and fix a curve $e \subset T_1$ contracted by f . For each j , let D_j be a rational Cartier divisor on T , and let $k_j > 0$ be an integer such that $k_j D_j$ is Cartier. Suppose a nonzero linear system of $k_j f^* D_j$ is generated outside the inverse image of a finite subset of T . Let F_j be its fixed divisor. If, for one fixed $c > 0$,*

$$\text{coeff}_e F_j \geq ck_j \quad \text{for all } j,$$

then there exists $\epsilon > 0$, independent of j , such that

$$(3.10) \quad D_j^2 \cdot [T] \geq \epsilon \quad \text{for all } j.$$

No normality assumption on T is needed.

Proof. Every curve in $\text{Supp } F_j$ maps to a point of T , by the generation assumption. Fix a very ample Cartier divisor H_T on T and an ample Cartier divisor J on T_1 , and put $Q = f^* H_T$. The projection formula gives

$$(3.11) \quad Q^2 = H_T^2 \cdot [T] > 0, \quad Q \cdot F_j = f^* D_j \cdot F_j = 0.$$

The Hodge index theorem on the fixed smooth surface T_1 [Har77, Chapter V, Section 1] makes the negative intersection form positive definite on $Q^\perp \subset N^1(T_1)_\mathbb{R}$. Consequently there is one constant $K > 0$ such that

$$(J \cdot F)^2 \leq K(-F^2) \quad \text{for every } F \in Q^\perp.$$

Effectiveness of F_j gives $J \cdot F_j \geq ck_j(J \cdot e)$. Thus, with the fixed positive number $\epsilon = c^2(J \cdot e)^2/K$,

$$(3.12) \quad F_j^2 \leq -\epsilon k_j^2.$$

After removal of F_j , the residual linear system has no fixed curve. Its square is nonnegative: choose two members with no common curve and take their effective intersection, or use a nowhere vanishing member if the system is trivial. By (3.11) and the projection formula,

$$0 \leq (k_j f^* D_j - F_j)^2 = k_j^2 (D_j^2 \cdot [T]) + F_j^2.$$

Now (3.12) proves (3.10). □

3.4. The nefness criterion. We now combine the last two estimates. A negative curve forces all sections to vanish linearly at one fixed ambient point. Restriction to a surface through that point, followed by one blowup, produces the fixed exceptional curve required by Lemma 3.3. The choices are made before the perturbation index varies.

Proposition 3.4 (Vanishing surface square implies nefness). *Assume (3.1). If, for one very ample Cartier divisor H on Y ,*

$$(3.13) \quad L^2 \cdot H^{n-2} = 0,$$

then L is nef.

Proof. Suppose that an integral curve $C_0 \subset Y$ satisfies $L \cdot C_0 < 0$. Fix a projective resolution $\pi : \hat{Y} \rightarrow Y$ which is an isomorphism over the smooth locus. Choose a closed point $x \in C_0$ outside the images of those irreducible components of $\pi^{-1}(C_0)$ which do not dominate C_0 . There are only finitely many such images, and each is a point. Consequently every point of $\pi^{-1}(x)$ lies on a component of $\pi^{-1}(C_0)$ dominating C_0 .

A surface through x . Choose $n - 2$ very general members $H_1, \dots, H_{n-2} \in |H|$, all passing through x . They meet properly, and their intersection meets each Z_j and $\text{Sing } Y$ in at most finitely many points. Indeed, the hyperplanes through x have no other base point; successive general cuts avoid

all positive-dimensional components of the relevant intersections. The countably many conditions can again be imposed simultaneously over $\mathbb{C}$. Write their intersection cycle as

$$H_1 \cdots H_{n-2} \cdot [Y] = \sum_{i=1}^a a_i [T^{(i)}], \quad a_i > 0,$$

where each $T^{(i)}$ is an integral surface. The dimension theorem guarantees a component through x ; fix one and denote it by T .

On every $T^{(i)}$, the system induced by $|d_j M_j|$ has at most a finite base locus. Its square is nonnegative, including when $T^{(i)}$ is nonnormal. To check this directly, choose a section nonzero at the generic point, and then choose a second section not vanishing identically on any curve of the first zero divisor. The resulting Cartier intersections with $[T^{(i)}]$ form an effective zero-cycle. If the first section has no zero, the restricted line bundle is trivial and its square is zero. These are the usual Cartier intersection and projection formulas on cycles [Ful98, Chapter 2]. It follows that

$$M_j^2 \cdot [T^{(i)}] \geq 0, \quad \sum_{i=1}^a a_i (M_j^2 \cdot [T^{(i)}]) = M_j^2 \cdot H^{n-2} \rightarrow 0$$

by (3.13). In particular,

$$(3.14) \quad M_j^2 \cdot [T] \rightarrow 0.$$

Fixed geometry above x . The surface T meets $\text{Sing } Y$ only in finitely many points. Its strict transform $T_{\hat{Y}}$ is thus birational to T and maps onto it. Choose any $z \in T_{\hat{Y}}$ above x . There is an integral curve $\Gamma \subset \hat{Y}$ through z dominating C_0 . To see this, take an irreducible component $A \subset \pi^{-1}(C_0)$ through z . By the choice of x , it dominates C_0 . If $\dim A = d$, its fiber A_x is a proper closed subset of dimension at most $d - 1$. General very ample cuts through z , in number $d - 1$, cut A_x to dimension at most zero and leave a curve component through z on A . That component cannot lie in A_x , so it dominates C_0 . For $d = 1$, take $\Gamma = A$.

Let $\iota : N \rightarrow \hat{Y}$ denote the normalization map of Γ , followed by inclusion. Resolve $T_{\hat{Y}}$ by a smooth projective surface $T_0 \rightarrow T_{\hat{Y}}$, choose a point $w \in T_0$ above z , and blow it up. Write T_1 for the resulting surface, $e \subset T_1$ for the exceptional curve of this blowup, and $f : T_1 \rightarrow T$ for the composite map:

$$\begin{array}{ccccccc} T_1 & \xrightarrow{\text{Bl}_w} & T_0 & \longrightarrow & \hat{Y} & \xleftarrow{\iota} & N \\ f \downarrow & & & & \downarrow \pi & & \downarrow \\ T & \longleftarrow & & & Y & \longleftarrow & C_0 \end{array}$$

The images of T_0 and N in $\hat{Y}$ meet at z . The exceptional curve e , all these varieties, and all these maps are fixed throughout the remaining argument.

The contradiction on the fixed surface. The projection formula and $M_j \rightarrow L$ give, for one fixed $\delta > 0$,

$$(3.15) \quad \deg_N \iota^* \pi^* M_j \leq -\delta \quad \text{for all sufficiently large } j.$$

In fact this degree equals the positive degree of $N \rightarrow C_0$ times $M_j \cdot C_0$. For each such j , choose any positive multiple k_j of d_j . Lemma 3.2, applied on the fixed $\hat{Y}$ and Γ , shows that every nonzero pullback section $s' \in H^0(\hat{Y}, k_j \pi^* M_j)$ of a section on Y satisfies

$$(3.16) \quad \text{ord}_z(s') \geq c k_j, \quad c = \delta/g > 0.$$

The constant c depends only on the fixed normalized curve and the negative-degree margin in (3.15).

The sections on Y induce a nonzero linear system of $k_j f^*(M_j|_T)$, generated outside $f^{-1}(T \cap Z_j)$. Nonzeroness follows from generation at points of $T \setminus Z_j$.

Every nonzero section in this induced system has order at least ck_j at w before the last blowup. Indeed, the local ring map $\mathcal{O}_{\widehat{Y},z} \rightarrow \mathcal{O}_{T_0,w}$ sends $\mathfrak{m}_z^b$ into $\mathfrak{m}_w^b$; apply (3.16) to a section inducing it. Sections whose restriction is identically zero do not contribute to the induced system. Thus its fixed divisor F_j satisfies $\text{coeff}_e F_j \geq ck_j$. All the hypotheses of Lemma 3.3 hold with $D_j = M_j|_T$. Lemma 3.3 gives a fixed $\epsilon > 0$ such that

$$M_j^2 \cdot [T] \geq \epsilon \quad \text{for all sufficiently large } j,$$

contrary to (3.14). Therefore no curve C_0 of negative L -degree exists, and L is nef. $\square$

4. THE CANONICAL DIVISOR

We now apply the preceding results to the canonical divisor. The passage back to the original smooth variety is recorded explicitly, because the numerical-dimension hypothesis concerns a fixed ample twist on X .

Proof of Theorem 1.1. Apply Proposition 2.1 to obtain a finite canonical MMP

$$X \dashrightarrow Y$$

and the divisors $L = K_Y$, B_Y , and $M_j = L + t_j B_Y$, with generated multiples outside closed subsets Z_j of codimension at least three. The variety Y is terminal and therefore smooth in codimension two. These are the standing hypotheses of the numerical results.

Fix a very ample Cartier divisor H on Y and a projective resolution $\pi : \widehat{Y} \rightarrow Y$ that is an isomorphism over the smooth locus. Lemma 3.1 gives a fixed Cartier divisor P on $\widehat{Y}$. Choosing one positive integer r_0 for which $r_0 L$ is Cartier, we obtain

$$(4.1) \quad h^0(\widehat{Y}, m\pi^*L + P) \geq \frac{m^2}{2} L^2 \cdot H^{n-2} + O(m) \quad (r_0 \mid m).$$

It also gives $L^2 \cdot H^{n-2} \geq 0$.

Suppose that this intersection number is positive. Choose a smooth projective common resolution W dominating $\widehat{Y}$ and all the models in the finite MMP prefix, with maps

$$r : W \longrightarrow \widehat{Y}, \quad u : W \longrightarrow X.$$

Proposition 2.1 gives

$$u^*K_X - r^*\pi^*L \geq 0.$$

For every positive multiple m of r_0 , this effective difference gives an injection

$$H^0(\widehat{Y}, m\pi^*L + P) \hookrightarrow H^0(W, mu^*K_X + r^*P).$$

Set $P_X = u_*r^*P$. This is a fixed integral Weil divisor, hence Cartier on the smooth X . A rational section f in the right-hand space satisfies

$$\text{div}_W(f) + mu^*K_X + r^*P \geq 0.$$

Pushing this inequality forward gives

$$\text{div}_X(f) + mK_X + P_X \geq 0.$$

Thus there is an injective linear map

$$H^0(W, mu^*K_X + r^*P) \hookrightarrow H^0(X, mK_X + P_X).$$

Choose a single ample Cartier divisor A_X for which $A_X - P_X$ has a nonzero section. Multiplication by that fixed section gives

$$h^0(X, mK_X + A_X) \geq h^0(\widehat{Y}, m\pi^*L + P).$$

By (4.1), the left-hand side is bounded below by cm^2 , for a fixed $c > 0$, along all sufficiently large multiples of r_0 . Consequently

$$\limsup_{m \rightarrow \infty} \frac{h^0(X, mK_X + A_X)}{m^2} > 0,$$

contradicting $\kappa_\sigma(X, K_X) = 1$ and the convention (1.1). It follows that

$$L^2 \cdot H^{n-2} = 0.$$

Proposition 3.4 now implies that $L = K_Y$ is nef, including on curves contained in the singular locus. All other conclusions of Theorem 1.1, including the effective exceptional comparison (1.2) and its strict support condition, are supplied by the finite MMP in Proposition 2.1. $\square$

REFERENCES

- [AHK07] Valery Alexeev, Christopher D. Hacon, and Yujiro Kawamata, *Termination of (many) 4-dimensional log flips*, *Inventiones Mathematicae* **168** (2007), no. 2, 433–448.
- [BCHM10] Caucher Birkar, Paolo Cascini, Christopher D. Hacon, and James McKernan, *Existence of minimal models for varieties of log general type*, *Journal of the American Mathematical Society* **23** (2010), no. 2, 405–468.
- [Bir10] Caucher Birkar, *On existence of log minimal models*, *Compositio Mathematica* **146** (2010), no. 4, 919–928.
- [Dru11] Stéphane Druel, *Quelques remarques sur la décomposition de Zariski divisorielle sur les variétés dont la première classe de Chern est nulle*, *Mathematische Zeitschrift* **267** (2011), no. 1–2, 413–423.
- [Ful98] William Fulton, *Intersection theory*, second ed., *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*, vol. 2, Springer-Verlag, Berlin, 1998.
- [Gon11] Yoshinori Gongyo, *On the minimal model theory for dlt pairs of numerical log Kodaira dimension zero*, *Mathematical Research Letters* **18** (2011), no. 5, 991–1000.
- [Har77] Robin Hartshorne, *Algebraic geometry*, *Graduate Texts in Mathematics*, vol. 52, Springer-Verlag, New York, 1977.
- [Hir64] Heisuke Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II*, *Annals of Mathematics* **79** (1964), 109–203, 205–326.
- [KM98] János Kollár and Shigefumi Mori, *Birational geometry of algebraic varieties*, *Cambridge Tracts in Mathematics*, vol. 134, Cambridge University Press, Cambridge, 1998.
- [Laz04a] Robert Lazarsfeld, *Positivity in algebraic geometry I: Classical setting, line bundles and linear series*, *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*, vol. 48, Springer-Verlag, Berlin, 2004.
- [Laz04b] ———, *Positivity in algebraic geometry II: Positivity for vector bundles, and multiplier ideals*, *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*, vol. 49, Springer-Verlag, Berlin, 2004.
- [Laz10] ———, *A short course on multiplier ideals*, *Analytic and Algebraic Geometry*, IAS/Park City Mathematics Series, vol. 17, American Mathematical Society, Providence, RI, 2010, pp. 451–494.
- [LP18] Vladimir Lazić and Thomas Peternell, *Abundance for varieties with many differential forms*, *Épjournal de Géométrie Algébrique* **2** (2018), 1–35, Article 1.
- [LX25] Jihao Liu and Zheng Xu, *Non-vanishing implies numerical dimension one abundance*, 2025, arXiv:2505.05250v2, 30 July 2025.
- [Mor88] Shigefumi Mori, *Flip theorem and the existence of minimal models for 3-folds*, *Journal of the American Mathematical Society* **1** (1988), no. 1, 117–253.
- [Nad90] Alan Michael Nadel, *Multiplier ideal sheaves and Kähler–Einstein metrics of positive scalar curvature*, *Annals of Mathematics* **132** (1990), no. 3, 549–596.
- [Nak04] Noboru Nakayama, *Zariski-decomposition and abundance*, *MSJ Memoirs*, vol. 14, Mathematical Society of Japan, Tokyo, 2004.
- [Ope26] OpenAI, *Log abundance in characteristic zero*, OpenAI Math Release preprint [OAI:Log-abundance-in-characteristic-zero-September-24-2026](#), 2026, Theorem 1.1.